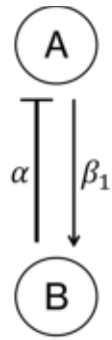


A Integral feedback model

$$\frac{dA}{dt} = \alpha(\mu - B)$$

$$\frac{dB}{dt} = \beta_1 A - \beta_2 B$$

$$B_{ss} = \mu$$



Steady-state of B depends only on μ , and is thus robust to β_1 and β_2 .

C Integral feedback model is equivalent to a mass on a spring

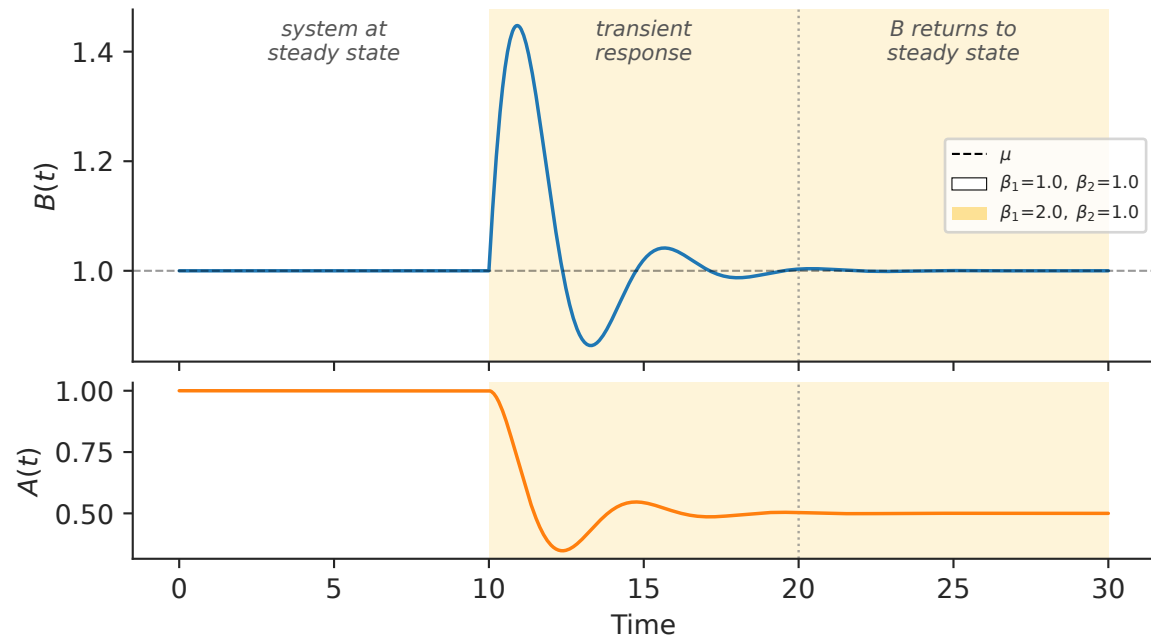
$$\frac{d^2 B}{dt^2} + \beta_2 \frac{dB}{dt} + \alpha \beta_1 B = \alpha \beta_1 \mu$$

mass $m = 1$

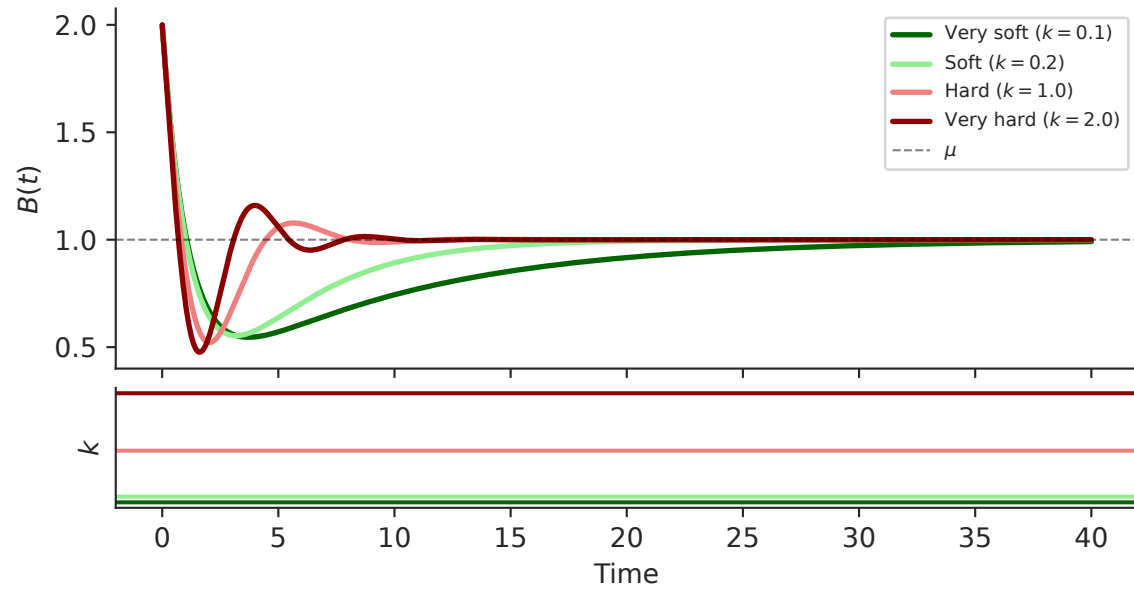
damping $c = \beta_2$

spring constant $k = \alpha \beta_1$

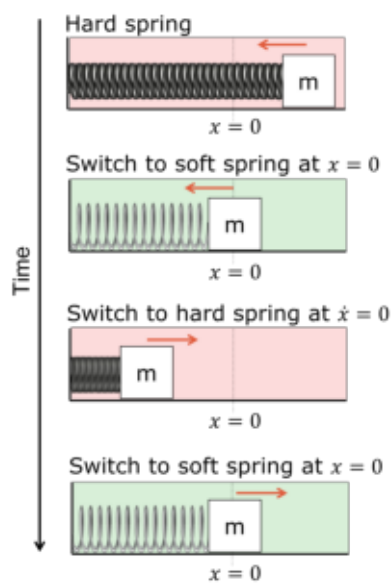
B Integral feedback ensures B converges to μ after parameter change



D B converges to μ for any fixed spring constant



E Adaptive strategy that amplifies the motion of a mass on a spring



G Calcium model with integral feedback is also equivalent to a mass on a spring

$$P = k_0(R_0 - Ca)$$

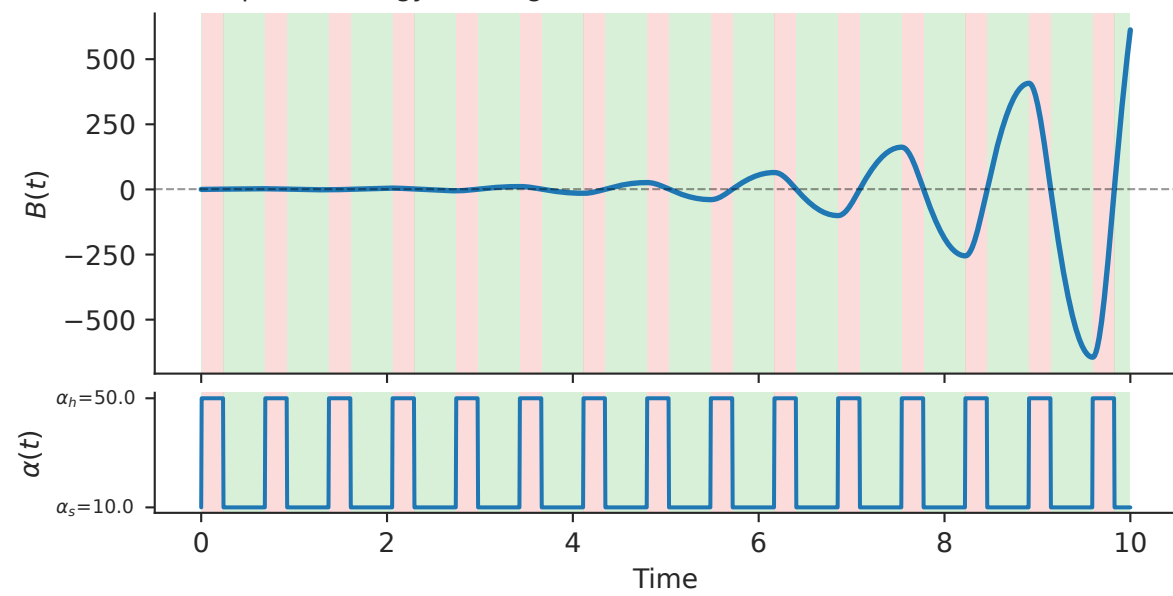
$$\dot{Ca} = k_1 P + k_2 D$$

$$\dot{D} = k_3 P$$

$$Ca_{ss} = R_0$$

spring constant $k = k_2 k_3 k_0$

F Adaptive strategy drives growth in AB model



H Adaptive strategy drives growth in calcium model

